

Shrey Anand

onlyshrey98@gmail.com | +91-8340463043 | GitHub | LinkedIn | Portfolio

EDUCATION

Vellore Institute of Technology, Chennai

2023 – 2027

B.Tech, Computer Science and Engineering (AI & ML) | CGPA: 8.46

TECHNICAL SKILLS

Languages: Java, Python, C++, SQL

Backend: Spring Boot, Spring MVC, Spring Data JPA, RESTful APIs, gRPC, Microservices

Databases: PostgreSQL, MySQL, MongoDB, Redis

Cloud / DevOps: AWS (EC2, S3), Docker, GitHub Actions, Nginx, Git

AI Integration: LLM APIs, RAG (LangChain), Agentic tool orchestration (LangGraph)

EXPERIENCE

HCLTech — Software Engineering Intern

May 2026 – July 2026

- migrated legacy
- Validated all ported libraries across Linux VMs and production hardware using automated C# test harnesses (Mono P/Invoke), confirming behavioral parity with the original Windows DLLs.

MedxAi Innovations Pvt. Ltd. — Summer Intern

May 2025 – August 2025

- Developed a full-stack microservices-based cervical-cancer screening platform using Spring Boot with REST APIs for image upload, result retrieval, and clinician workflows, integrating WebSockets for real-time notifications and WebRTC for live video transmission during remote screenings.
- Implemented JWT-based role-based auth for doctors and admin staff with HIPAA-compliant access controls, encrypted storage, and audit logging.
- Deployed microservices on AWS ECS (Fargate) with GitHub Actions CI/CD, reducing deployment time by 80%.

Samsung PRISM — Research Intern

Aug 2025 – June 2026

- Developed Python-based data preprocessing and evaluation pipelines for Test-Time Adaptation (TTA) speech recognition, automating dataset preparation and Word Error Rate (WER) benchmarking across noisy acoustic datasets.

PROJECTS

StreamForge — Distributed Media Processing & Streaming System

GitHub | Java, Spring Boot, Kafka, PostgreSQL, Redis, Docker

- Built a distributed media platform using Spring Boot microservices, Eureka service discovery, and an API Gateway, with database-per-service PostgreSQL isolation, Redis caching, and MinIO object storage, all orchestrated via Docker Compose.
- Built an event-driven video-ingestion pipeline using Kafka that stores raw uploads in object storage and processes them asynchronously via worker services with idempotent job tracking and dead-letter-queue handling.
- Implemented FFmpeg-based media processing to extract metadata, generate thumbnails, and transcode uploads into multi-resolution HLS variants, served through a backend HLS proxy for adaptive-bitrate streaming.

MedSync — Distributed Healthcare Microservices Platform

GitHub | Live | Java, Spring Boot, Kafka, PostgreSQL, LangChain, LangGraph, Docker

- Built a distributed healthcare backend with 6 Spring Boot microservices (isolated PostgreSQL databases), connecting billing services via gRPC and streaming events through Kafka + Protobuf for analytics, all routed through a JWT-secured API Gateway.
- Built an AI operations agent (FastAPI + LangGraph, ReAct pattern) with 20+ callable tools that autonomously chains patient registration, billing, and scheduling calls across microservices from a single natural-language request.
- Deployed the platform on AWS EC2 using Docker Compose (11 containers) behind an Nginx reverse proxy (TLS 1.3), served the frontend via S3 + CloudFront, and automated releases with GitHub Actions CI/CD including health checks and automatic rollback on failure.

ACHIEVEMENTS

- Co-authored “Ship Hull Defect Detection and Instance Segmentation with a ShuffleBlock Backbone and Texture-Aware Attention,” published in *Results in Engineering* (Elsevier, Impact Factor 7.9), achieving 89.74% IoU — DOI.
- Solved 350+ Data Structures Algorithms problems across LeetCode and GeeksforGeeks.

EXTRACURRICULARS

- Organized and led a hands-on technical workshop on building LLM-powered applications for 30+ participants as Student Coordinator, managing logistics and mentoring hands-on sessions.
- Represented MedxAi Innovations Pvt. Ltd at UImagineTN 2026 Tech summit, engaging with founders and delegates, interacting with visitors and students, providing them insights on applications of Artificial Intelligence in Healthcare